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Sustainable fishing of a protogynous species: The Ballan Wrasse (*Labrus bergylta*)

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Abstract

The Ballan Wrasse is subject to a live fishery in Scotland targeted at providing cleaner fish that eat sea lice for salmon aquaculture. As it is a protogynous species, with populations presenting a skewed sex ratio and bigger males than females, concerns concerning its vulnerability to overfishing have been raised and minimum and maximum size limits were deployed. As an attempt to evaluate the species vulnerability to overfishing along Scottish coasts, I modelled a Ballan wrasse stock using an age-class model adapted to the specificities of the species, and data taken from the literature. Exposing the modelled stock to different fishing mortality rates and size limits highlighted that the stock was at risk of local extinction under the current Scottish and other European size limits at moderate to high fishing rates. These results suggest that further regulations might be necessary to prevent a Ballan wrasse stock extinction on Scottish coasts.

Keywords: overfishing vulnerability, hermaphrodite, age-class model, live fishery, sea lice, cleaner fish

1. Introduction

The Ballan wrasse (*Labrus bergylta*) is a fish that lives in the eastern Atlantic's subtidal and tidal zones. This fish inhabits rocky, algae-filled areas at depths ranging from 1 to 50 meters and feeds on mollusks and crustaceans (Dipper et al. 1977). Individuals have been reported to live for up to 34 years, measure as much as 66cm, and weigh as much as 4.35kg (IGFA 2001, Villegas-Rios et al. 2021, Pritchard et al. 2024). The Ballan wrasse forms harems, with each male defending a territory and fertilizing the eggs in the nests within it (Villegas-Rios et al. 2013). It is a protogynous sequential hermaphrodite, meaning all individuals are born female and transition to male during their lifetime under certain social conditions. These conditions are suspected to relate to the presence of males in the harem, the size of other fish in the population, and the size and dominance of the fish in question (Dipper et al. 1977, Munday et al. 2006). This sex change is associated with males being larger, and fewer males than females in a given population, with around 10% males (Dipper et al. 1977, Leclerq et al. 2014).

In recent years, the Ballan wrasse has become a popular cleaner fish for salmon aquaculture as it can feed on sea lice (Leclerq et al. 2013). In Scotland, captive breeding for the Ballan wrasse has been developed to fulfill the needs for salmon fisheries, but while production allows for covering a large part of the needs, some individuals used as cleaner fish are still commercially harvested from the wild in some locations (Scottish government. 2024a, Scottish government. 2024b). As the species' popularity for such uses has increased, concerns about overfishing in European waters have been raised, and varying minimum and maximum size limits have been implemented throughout Europe for commercial fishers to reduce the risk of local extinction in multiple locations (Pritchard et al. 2025a, Pritchard et al. 2025b).

In this context, this study aims to evaluate the current European minimum and maximum size limits on a Scottish Ballan wrasse population. I will assess the impact of these size limits on a simulated population's yield and population size under varying fishing rates. I will simulate the population using an age-class model.

2. Materials and methods

2.1. Model description

I simulated the population using an age-class model and a discrete time scale, with each class starting on the 1st of each year and ending on the 31st of December. Males and females were simulated separately to account for skewed sex ratios and sex change. Individuals are assumed to be all born females, live up to a maximum of 30 years old, and to reach sexual maturity at around 5 years old. Sex change is assumed to occur only after maturity (Leclerq et al. 2014). Effectives and other values are calculated as of the day before cohorts age by one year, on the 31st of December.

2.2. Model equations

2.2.1. Mortality

Fish numbers in an age class *a* at a given time *t* is a function of mortality and recruitment to fishery of the fish of the previous age class *a-1* of the precedent year *t-1.* It is described as:

$$N_{a+1,t+1} = N_{a,t}e^{-Z_a},$$

Where Z_a is the total instantaneous mortality rate at age. Z_a can be calculated as the sum of the natural mortality M and the fishing mortality rate F_a.

2.2.2. Growth

Length-at-age (L_a) was calculated using Von Bertalanffy's growth equation:

$$L_a = L_\infty\left(1 - e^{-k(a-t_0)}\right),$$

Where L_∞ is the asymptotic length of the fish, k is the growth rate, and t_0 is the theoretical length-at-age 0. Mass (m) was calculated using a simple allometric mass-length relationship:

$$m = \alpha_0 L^{\alpha_1},$$

Where α_0 and α_1 are parameters that define the mass-length relationship.

2.2.3. Sex change

The probability of sex change ($P_c(L)$) was calculated as a function of length, following the relative size rule of Alonzo & Mangel (2005). Following this rule, the mean size of all individuals in the population determines the probability of an individual undergoing sex change:

$$P_c(L) = \frac{1}{1 + \exp\left(-b\left(L - (L_i + \Delta L_c)\right)\right)},$$

Where L_i is the average size in the population, ΔL_c is the difference above the mean at which the probability of undergoing sex change equals 0.5, and b is a shape parameter. Transitioning individuals are integrated into the equation of the count of individuals on time and mortality separately for both females and males. The equation for females deduces the transitioning individuals:

$$N_{F,a+1,t+1} = (N_{F,a,t} - (N_{F,a,t} \times P_c(L_{aF})))e^{-Z_a},$$

And the equation for males introduces the transitioning individuals:

$$N_{M,a+1,t+1} = (N_{M,a,t} + (N_{F,a,t} \times P_c(L_{aF})))e^{-Z_a}.$$

2.2.4. Fecundity

The egg count (E) was calculated for mature females as a function of length, as shown in the work of Villegas-Rios et al. (2014):

$$E = e^{\beta_0 L + \beta_1},$$

Where β_0 and β_1 are parameters that define the egg count at length relationship.

2.2.5. Fertilization

As the number of males can become limiting for the stock recruitment in this model, I included a measure of fertilization rate ($f(x_F)$) for the fished population developed by Brooks et al. (2008):

$$f(x_F) = \frac{4kx_F}{(1-k) + (5k-1)x_F},$$

Where k is a measure of steepness and defines how much male depletion affects the fertilization rate, and x_F which is the ratio of the proportion of males in the population at a certain time to the proportion of males in the same stable and unfished population (under natural mortality only):

$$x_F = \frac{p_t}{p_{Z=M}}.$$

The number of fertilized eggs (ψ) was calculated from the total number of eggs in the population and the fertilization rate:

$$\psi = f(x_F)E.$$

2.2.6. Stock recruitment

The recruitment to the age-class 0 ($R(\psi)$) was calculated using a modified version of Beverton-

Holt's spawner-recruit model developed by Brooks et al. (2008), which depends on the number of fertilized eggs (ψ):

$$R(\psi) = \frac{4hR_0\psi}{R_0\phi_0(1-h) + (5h-1)\psi},$$

Where ϕ_0 is the number of fertilized eggs per recruit in the unfished stable population, R_0 is the unfished recruitment, which can also be interpreted as the maximum population size, and h is a measure of steepness and is analogous to k in the fertilization equation.

2.2.7. Recruitment to fishing

The probability of a fish of age a to be recruited to fishing ($F_a(L_a)$) is determined as:

$$F_a(L_a) = \frac{1}{1+e^{\left(-\frac{L_a - L_{min}}{\eta}\right)}} * \left(1 - \left(\frac{1}{1+e^{\left(-\frac{L_a - L_{max}}{\eta}\right)}}\right)\right) F,$$

Where L_{min} is the minimum size limit, L_{max} is the maximum size limit, and η is a parameter which defines how precise the size limits are. The recruitment to fishing is integrated into the total mortality at age (Z_a).

$$Z_a = M + F_a$$

2.2.8. Calculation of catches and yield

Catches (C_t) and yield (Y_t) where calculated using Baranov's catch equation modified to account for the different characteristics of males and females:

$$C_t = \sum_{a=0}^{a=30} \frac{F_a}{Z_a}(1-e^{-Z_a})N_{F,a-1,t-1} + \sum_{a=0}^{a=30} \frac{F_a}{Z_a}(1-e^{-Z_a})N_{M,a-1,t-1},$$

$$Y_t = \sum_{a=0}^{a=30} \frac{F_a}{Z_a}(1-e^{-Z_a})N_{F,a-1,t-1}\, m_{F,a}/1000 + \sum_{a=0}^{a=30} \frac{F_a}{Z_a}(1-e^{-Z_a})N_{M,a-1,t-1}\, m_{F,a}/1000$$

2.3. Model parametrization

Parameters were taken from literature when available, and were otherwise assumed or chosen through iteration to reproduce real population data as closely as possible (Table 1).

Table 1: Model parameters

Parameter name	**Parameter symbol**	**Parameter value**	**Source**
Initial population size	N_0	30	Chosen
Natural mortality rate	M	0.2	Assumed
Von Bertalanffy's asymptotic body length females	$L_{\infty F}$	354.4	Leclerq et al. 2014
Von Bertalanffy's growth coefficient females	k_F	0.214	Leclerq et al. 2014
Von Bertalanffy's initial age females	t_{0F}	-0.98	Leclerq et al. 2014
Von Bertalanffy's asymptotic body length males	$L_{\infty M}$	366.6	Leclerq et al. 2014
Von Bertalanffy's growth coefficient males	k_M	0.244	Leclerq et al. 2014
Von Bertalanffy's initial age males	t_{0M}	-3.19	Leclerq et al. 2014
Body allometry constant	α_0	0.00931	Leclerq et al. 2014
Body allometry exponent	α_1	3.146	Leclerq et al. 2014
Sex change shape parameter	b	0.1	Estimation through iteration
Difference from the mean size at which $P_c(L) = 0.5$	ΔL_c	115	Estimation through iteration
Fecundity multiplier	β_0	0.117	Villegas-Rios et al. 2014
Fecundity constant	β_1	9.563	Villegas-Rios et al. 2014
Fertilization steepness	k	0.5	Assumption
Unfished recruitment	R_0	10^6	Chosen
Stock recruitment steepness	h	0.4	Assumption
Size limit precision	η	10	assumption
Fishing mortality rate	F	Varied	
Minimum size limit	L_{min}	Varied	

Maximum size limit	L_{max}	Varied	

2.4. Model analysis

I simulated the population over 250 years without fishing to find a population equilibrium, then introduced fishing for 500 years. The values of the last year of the simulation were picked as the model result, as an equilibrium was reached.

The analysis aimed to observe the long-term consequences on yield and population abundance of commercial minimum and maximum size limits already implemented for this species throughout Europe on our population, under a mortality rate F varying from 0 to 1 with a step of 0.05. I additionally simulated the consequences of higher size limits corresponding to theoretically sustainable practices, adapted so that individuals of higher sizes are conserved (Table 2) (Beverton & Holt. 1957).

Table 2: Size limits (LS) tested on this model (mm). modified after Pritchard et al. 2025a

Location	**Minimum LS**	**Maximum LS**
England, Devon and Severn IFCA	180	260
Norway	220	280
Scotland	120	240
Sweden	150	300
No location, Theoretically sustainable	290	350

3. Results

3.1. Effect of size limits and fishing mortality rate on population size

All European size limits lead to a population crash below the fishing mortality rate of 0.45 after 500 years. The crash happened at a lower fishing mortality rate under Swedish regulations (150-300mm), then Scottish regulations (120-240mm), English regulations (180-260mm), and lastly Norwegian regulations (220-280mm). The population size decreased under theoretically sustainable size limits, but did not crash under a fishing mortality rate up to 1 (Figure 1).

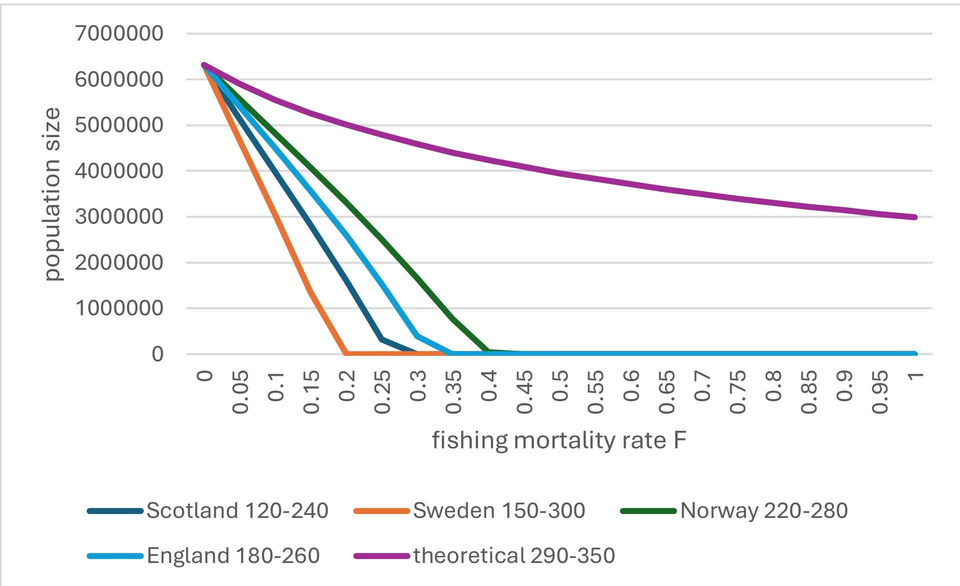

Figure 1: population size under different size limits as function of fishing mortality rate

3.2. Effect of size limits and fishing mortality rate on yield

A maximum yield was not achieved under the theoretically sustainable size limit but was achieved under other size limits. The highest yield was achieved with the theoretically sustainable yield. Other size limits had lower maximum yields the smaller the fishing pressure under which the population crashed.

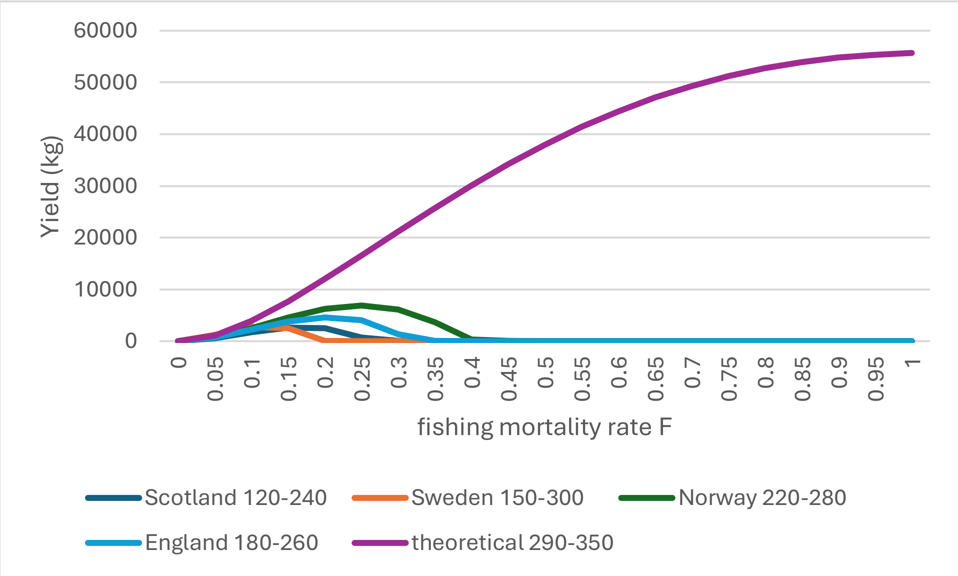

Figure 2: yield under different size limits as function of fishing mortality rate

3.3. Effect of size limits and mortality rate on catch

The number of fish caught was higher with the European size limits, up to a fishing rate of between 0.2 and 0.25, after which the value is higher with the theoretically sustainable size limits. For the European size limits, the catch drops to zero, respectively to the fishing rate at which the population crashes.

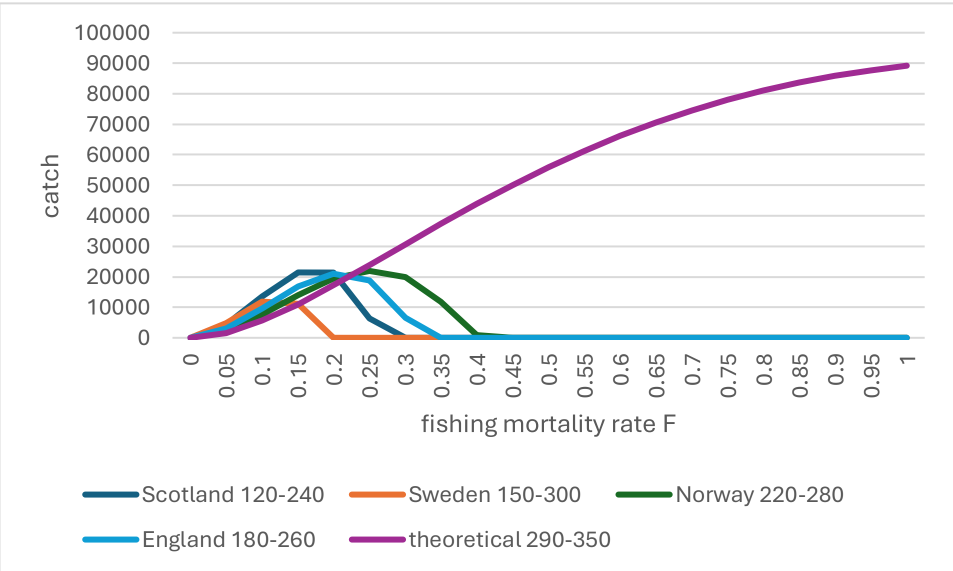

Figure 3: catch under different size limits as function of fishing mortality rate

4. Discussion

The results indicate that current Scottish minimum and maximum size limits would lead to population extinction in the long term under moderate to high fishing pressures. Using the theoretically sustainable higher size limits that allow individuals to reach maturity and reproduce before being recruited to the fishery, and that allow to avoid the recruitment of bigger individuals which are likely to be males, would prevent the population from crashing under a fishing pressure higher than 0.2-0.4, unlike current measures. Higher size limits would also permit a higher yield in all circumstances, and a higher catch under fishing pressures superior to 0.25.

Interestingly, current fishing regulations at low fishing pressures (0.05-0.2) provide a higher catch but a lower yield than the theoretically sustainable size limits. These results might reflect that the purpose of the current size limits is not to provide a higher yield, but a higher number of young fish, if current fishing pressures remain low, as would indicate the sustained landings for this fishery over the years (Scottish government. 2024a) .

The fact that the current commercial size limits are tailored to prioritize the capture of a high number of young fish makes sense commercially, as the Ballan wrasse is fished for the purpose of eating sea lice in reared salmon enclosures (Leclerq et al. 2013). Contrary to classic fisheries for which the value is weight-wise, the Ballan wrasse's value probably lies in their number and age, as bigger fish might not survive as long in salmon enclosures and therefore might be less efficient relative to its price.

The fact that the population is at risk of local extinction under the current European fishing size limits at moderate and high fishing mortality rates is concerning, as the fishing pressure could easily increase to these levels for this fishery. These findings align with the results presented by Pritchard et al. (2025b), who classified the Ballan wrasse as a moderately vulnerable species to extinction under the current fishing pressure. Therefore, further measures to either control the fishing mortality rate or increase the capture size of the fish seem necessary to avoid a potential stock crash.

These results should be considered with caution as the model used presents a few flaws. Firstly, I did not incorporate a maturity ogive and instead considered all individuals to be mature at age 5. Therefore, it would be possible for females to mature earlier or later than is this model, and males of age 5 could therefore appear in the population, which is not the case in this model. A consequence of this is that it could be possible for fishing size limits to be less harmful for the stock as younger and smaller individuals might be mature or change sex than in our model. Secondly, this model does not take into account environmental variations like temperature or the presence of prey and predators, which could have great implications for the stock. As environmental variations might have positive or negative effects on the stock, situations with low population effectives should be avoided, as instability in the environmental factors combined with a higher fishing mortality rate could cause a stock extinction. Thirdly, as individuals transition into males, their length takes the value as for males in the population and therefore experience an unrealistic shift in length. As size is also the basis for sex change calculation in our model, it would have been preferable to build a size-structured model, which would have allowed for more realistic dynamics in the stock.

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